

Komal Kale

Pune, Maharashtra

+91-6355804179

kalekomal2227@gmail.com

GitHub

LinkedIn

Career Objective

Passionate **BCA student** with a strong interest in **data analytics and web development**. Knowledgeable in **Python, SQL, and Excel**, with hands-on experience in data manipulation and visualization. Proficient in **HTML, CSS, and JavaScript** for building responsive, data-driven web applications. Always eager to learn, build innovative solutions, and continuously improve technical skills.

Education

JSPM University, Pune

Jul 2023 – May 2026

Bachelor of Computer Applications (BCA)

Chanakya Vidhyalaya, Ankleshwar, Gujarat

Jul 2020 – Jun 2023

Higher Secondary (12th)

Internship

Python Development Intern – Cognifyz Technologies (Remote)

Sep 2025 – Oct 2025

Virtual Internship

- Worked with Python and Pandas to clean and organize datasets for analysis.
- Performed basic data processing and preparation on structured text data.
- Created simple data visualizations using Matplotlib to understand trends and patterns.
- Gained hands-on experience in data handling, analysis, and Python-based workflows.

Training Programs

- **Employability Skills Training Program (Python)** — Mahindra Pride Classroom *2025–2026*
- **Python Developer For Future Skills And Youth Program** — Deutsche Bank in collaboration with Rubicon Foundation

Projects

Women Safety Application

- Developed a web-based platform connecting users and lawyers with structured onboarding and role-based access.
- Focused on data-driven features to improve accessibility, safety support, and user experience.

CSV Analyzer Application

- Built an application to analyze CSV files and extract meaningful insights using data processing techniques.
- Generated visual representations such as charts to support data interpretation and analysis.

Population Map Visualization

- Created a visualization tool to map population distribution by gender using CSV data and geo-coordinates.
- Displayed demographic insights through clear, interactive, and easy-to-understand maps.

TB Care:AI-Based Tuberculosis Detection and Healthcare Assistant

- Tuberculosis is a life-threatening infections disease affecting millions worldwide.Early detection through chest X-rays is critical but require expert radiologists.
- AI-powered system using Deep Convolutional Neural Network(CNNs) for automated TB detection and LLaMA-3 based conversational AI for explainable diagnosis support.

Technical Skills

- **Programming:** Python, SQL, JavaScript
- **Data Tools:** MS Excel , Pandas , PowerBI
- **Web Development:** HTML, CSS, Basic JavaScript
- **Database:** MySQL
- **Other Skills:** Data Cleaning , Logical Thinking, Problem Solving

Certifications

- Oracle Cloud Infrastructure 2025 DevOps Professional (1Z0-1109-25) — [View Credential](#)
- Oracle Autonomous Database Cloud 2025 Professional (1Z0-931-25) — [View Credential](#)
- Oracle Cloud Infrastructure 2025 AI Foundations Associate (1Z0-1122-25)
- Oracle Cloud Infrastructure 2025 Foundations Associate (1Z0-1085-25)
- Java Full Stack Developer Internship — [View Credential](#)
- Pregrad: Full Stack Development Program — [View Credential](#)

Publications

- **Komal Kale, K. Suraj, A. Kaldate, Y. Mahajan. "TB Care: AI-Based Tuberculosis Detection and Healthcare Assistant."** *Journal Publication of International Research for Engineering and Management (JOIREM)*, Vol. 04, Issue 4, April 2026. ISSN: 3107-6696.
 - Developed a custom CNN model achieving **93.0% validation accuracy** for Tuberculosis screening from chest X-rays, outperforming ResNet50 and MobileNetV2.
 - Implemented **Grad-CAM (Explainable AI)** to visually map and highlight pathologically infected lung areas to improve clinical trust.
 - Integrated a **LLaMA-3 driven conversational AI** to automate the generation of structured, human-readable medical reports and guide patient interaction.

Leadership & Achievements

- **Event Coordinator – JSPM University:** Conducted analytics and web development workshops, training 40+ students on visualization and data storytelling.
- **Hackathon Participant:** Developed a data-driven women safety platform; ranked among top 20 regional finalists.
- **Volunteer:** Guided peers on Python for data analysis and dashboard creation during university sessions.